George P. Kontoudis

Brown Hall BB 280C, 1500 Illinois St, Golden, CO 80401, USA
www.georgekontoudis.com ✉ george.kontoudis@mines.edu ◇ *Updated: August 6, 2026*

RESEARCH INTERESTS

My research interests lie in the intersection of **robotics**, **control theory**, and **machine learning**. What interests me most is the formulation of hybrid theoretical schemes and the development of novel robotic systems to bridge the gap between machine learning and autonomy. I am particularly interested in how learning algorithms and control theory can enable multi-agent systems to efficiently collaborate with minimal information exchange for robotic navigation and exploration tasks.

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Professor, Colorado School of Mines *Jan 2024–present*
Department of Mechanical Engineering

Core Faculty Member, Colorado School of Mines *Jan 2024–present*
Robotics Program

Affiliated Faculty Member, Colorado School of Mines *Aug 2025–present*
Quantum Engineering Program

Postdoctoral Research Associate, University of Maryland *Jan 2022–Dec 2023*
Motion and Teaming Lab (PI: Michael Otte)

Graduate Research Assistant, Virginia Tech *Aug 2016–Dec 2021*
Center for Marine Autonomy & Robotics (PI: Daniel J. Stilwell)
Computational Multiphysics Systems Laboratory (PI: Tomonari Furukawa)

Undergraduate Research Assistant, National Technical University of Athens *Apr 2014–Mar 2016*
Control Systems Laboratory (PI: Kostas J. Kyriakopoulos)

Founder & Research Associate, OpenBionics *Sep 2014–present*

EDUCATION

Ph.D. in Electrical Engineering, Virginia Tech *2018–2021*
Dissertation Title: "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration"

M.S. in Mechanical Engineering, Virginia Tech *2016–2018*
Thesis Title: "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation"

Diploma (B.S. & M.S.) in Mechanical Engineering, National Technical University of Athens *2012–2016*
Thesis Title: "Design and Development of an Underactuated, Anthropomorphic Robot Hand"

B.S. in Mechanical Engineering, University of West Attica *2005–2010*
Thesis Title: "Analysis of Compression and Air-Conditioning System in Short Takeoff and Landing Aircrafts"

TEACHING EXPERIENCE

Instructor, MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control *Fall '24, '25, '26*
Department of Mechanical Engineering, Colorado School of Mines
Participation - Evaluation: 29 - 3.49/5.00 (F '24), 25 - 4.17/5.00 (F '25), 33 (F '26)

Instructor, MEGN315 Dynamics *Fall '25, Spring '24, '25, '26*
Department of Mechanical Engineering, Colorado School of Mines
Participation - Evaluation: 31 - 4.04/5.00 (S '24), 37 - 4.06/5.00 (S '25), 47 - 3.59/5.00 (F '25), 51 - 4.08/5.00 (S '26)

Instructor, MEGN503 Graduate Seminar *Spring '25*
Department of Mechanical Engineering (Robotics and Biomechanics Division), Colorado School of Mines
Participation - Evaluation: 12 - 4.89/5.00 (S '25)

Co-Instructor, ENAE488O/788O Introduction to Autonomous Multi-Robot Swarms *Spring '23*
Department of Aerospace Engineering, University of Maryland

Graduate Teaching Assistant, ME4005, 4006 Mechancal Engineering Lab I, II *Fall '16, Spring '17*
Department of Mechanical Engineering, Virginia Tech

AWARDS & HONORS

NSF FRR-NRI Aspiring Robotics PI	*2023, 2024*
2022 Robotics: Science and Systems (RSS) Pioneer	*2022*
2022 MRC Postdoctoral Fellowship ($60K)	*2022*
5 × IEEE Student Travel Support for IROS, ACC, and CDC	*2015, 2019–2021*
2 × Virginia Tech GSA Travel Fund Award for Humanoids and ICORR	*2019, 2020*
NSF Student Travel Grant for WuWNet ($1K)	*2019*
NTUA Thomaideion Award for IROS	*2016*
Hackaday Prize, 2nd place among 900 projects ($10K)	*2015*
Robotdalen Innovation Award, 1st place (SEK 100K)	*2015*

INDUSTRIAL EXPERIENCE

Mechanical Engineer, Sychem S.A.	*Oct 2010–Aug 2015*
Aircraft Maintenance Engineer Trainee, Olympic Aviation	*May 2008–Jan 2010*

PUBLICATIONS

Underline denotes graduate student advisee co-author
$^{+}$ Plus denotes undergraduate student advisee co-author
* Asterisk denotes student mentee co-author

Journal Manuscripts Under Review (J:UR)

[1] Mykola Mazur, **George P. Kontoudis**, Michael F. Miller, Caroline R. Turner, Katherine R. Knaus, Anne K. Silverman, Anthony Petrella, "A Personalized Dual-IMU Wearable System for Ankle Joint Kinematics Estimation During Gait."

Referred Journal Publications (J)

[8] Josh Netter*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Decentralized Multi-Agent Motion Planning Using Cognitive Hierarchy and Gaussian Process Classification," *IEEE Transactions on Intelligent Vehicles*, 2026.

[7] **George P. Kontoudis**, Daniel J. Stilwell, "Multi-Agent Federated Learning using Covariance-Based Nearest Neighbor Gaussian Processes." *IEEE Transactions on Machine Learning in Communications and Networking*, 2025.

[6] **George P. Kontoudis**, Daniel J. Stilwell, "Scalable, Federated Gaussian Process Training for Decentralized Multi-Agent Systems," *IEEE Access*, 2024.

[5] Zirui Xu*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Online and Robust Intermittent Motion Planning in Dynamic and Changing Environments," *IEEE Transactions on Neural Networks and Learning Systems*, 2023.

[4] **George P. Kontoudis**, Stephen Krauss, Daniel J. Stilwell, "Model-Based Learning of Underwater Acoustic Communication Performance for Marine Robots," *Robotics and Autonomous Systems*, 2021.

[3] Geng Gao, Mojtaba Shahmohammadi, Lucas Gerez, **George P. Kontoudis**, Minas Liarokapis, "On Differential Mechanisms for Underactuated, Lightweight, Adaptive Prosthetic Hands," *Frontiers in Neurorobotics*, 2021.

[2] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Kinodynamic Motion Planning with Continuous-Time Q-Learning: An Online, Model-Free, and Safe Navigation Framework," *IEEE Transactions on Neural Networks and Learning Systems*, 2019.

[1] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, Tomonari Furukawa, "An Adaptive Actuation Mechanism for Anthropomorphic Robot Hands," *Frontiers in Robotics and AI*, 2019.

Chapters in Edited Volumes (BC)

[1] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, Zirui Xu*, "RRT-QX: Real-Time Kinodynamic Motion Planning in Dynamic Environments with Continuous-Time Reinforcement Learning," in *Brain and Cognitive Intelligence: Control in Robotics*, B. Wei (Ed.), Taylor & Francis Group, CRC Press, 2022.

Referred Conference Publications (C)

[21] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Bayesian Networks for Path-Based Sensors: Gathering Information and Path Planning in Communication Denied Environments," *World Symposium on the Algorithmic Foundations of Robotics (**WAFR**)*, Oulu, Finland, 2026.

[20] Meet Gandhi, **George P. Kontoudis**, "Distributed Quantum Gaussian Processes for Multi-Agent Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026.

[19] Sanket A. Salunkhe, **George P. Kontoudis**, "Federated Gaussian Process Learning via Pseudo-Representations for Large-Scale Multi-Robot Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026.

[18] Mykola Mazur, **George P. Kontoudis**, Anthony Petrella, "A Wearable Device for Accurate Ankle Movement Evaluation," *International Conference of the IEEE Engineering in Medicine and Biology Society (**EMBC**)*, Copenhagen, Denmark, 2025.

[17] **George P. Kontoudis**, Daniel J. Stilwell, "Decentralized Federated Learning using Gaussian Processes," *IEEE International Symposium on Multi-Robot and Multi-Agent Systems (**MRS**)*, Boston, USA, 2023.

[16] **George P. Kontoudis**, Michael Otte, "Adaptive Exploration-Exploitation Active Learning of Gaussian Processes," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Detroit, USA, 2023.

[15] **George P. Kontoudis**, Michael Otte, "Closed-Form Active Learning of Expected Variance Reduction using Gaussian Process Surrogates for Adaptive Sampling," *American Control Conference (**ACC**)*, San Diego, USA, 2023.

[14] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions," *International Symposium on Distributed Autonomous Robotic Systems (**DARS**)*, Montbéliard, France, 2022. *(Also appears as a chapter in the book: Distributed Autonomous Robotic Systems, Springer Tracts in Advanced Robotics, p. 286–299. 2024.)*

[13] Christos N. Mavridis, **George P. Kontoudis**, John S. Baras, "Sparse Gaussian Process Regression using Progressively Growing Learning Representations," *IEEE Conference on Decision and Control (**CDC**)*, Cancun, Mexico, 2022.

[12] Josh Netter*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Bounded Rational RRT-QX: Multi-Agent Motion Planning in Dynamic Human-Like Environments Using Cognitive Hierarchy and Q-Learning," *IEEE Conference on Decision and Control (**CDC**)*, Austin, USA, 2021.

[11] **George P. Kontoudis**, Daniel J. Stilwell, "Decentralized Nested Gaussian Processes for Multi-Robot Systems," *IEEE International Conference on Robotics and Automation (**ICRA**)*, Xi'an, China, 2021.

[10] Minas Liarokapis, **George P. Kontoudis**, "Teaching Robotic and Biomechatronic Concepts with a Gripper Design Project and a Grasping and Manipulation Competition," *IEEE International Conference on Robotics and Automation (**ICRA**)* , Xi'an, China, 2021.

[9] **George P. Kontoudis**, Daniel J. Stilwell, "Prediction of Acoustic Communication Performance in Marine Robots Using Model-Based Kriging," *American Control Conference (**ACC**)*, New Orleans, USA, 2021.

[8] Gal Gorjup, **George P. Kontoudis**, Anany Dwivedi, Geng Gao, Saori Matsunaga, Toshisada Mariyama, Bruce MacDonald, and Minas Liarokapis "Combining Programming by Demonstration with Path Optimization and Local Replanning to Facilitate the Execution of Assembly Tasks ," *IEEE International Conference on Systems, Man and Cybernetics (**SMC**)*, Toronto, Canada, 2020.

[7] **George P. Kontoudis**, Zirui Xu*, Kyriakos G. Vamvoudakis, "Online, Model-Free Motion Planning in Dynamic Environments: An Intermittent, Finite Horizon Approach with Continuous-Time Q-Learning," *American Control Conference (**ACC**)*, Denver, USA, 2020.

[6] **George P. Kontoudis**, Daniel J. Stilwell, "A Comparison of Kriging and Cokriging for Estimation of Underwater Acoustic Communication Performance," *ACM International Conference on Underwater Networks and Systems (**WuWNet**)*, Atlanta, USA, 2019.

[5] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, "An Adaptive, Humanlike Robot Hand with Selective Interdigitation: Towards Robust Grasping and Dexterous, In-Hand Manipulation," *IEEE-RAS International Conference on Humanoid Robots (**Humanoids**)*, Toronto, Canada, 2019.

[4] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, "A Compliant, Underactuated Finger for Anthropomorphic Hands," *IEEE/RAS-EMBS Inter. Conference on Rehabilitation Robotics (**ICORR**)*, Toronto, Canada, 2019.

[3] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Robust Kinodynamic Motion Planning using Model-Free Game-Theoretic Learning," *American Control Conference (**ACC**)*, Philadelphia, USA, 2019.

[2] Kyriakos D. Tsoukalas, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Active-Bayesian Learning for Cooperation Connectivity in Dynamic Cyber-Physical-Human Systems," *IEEE Symposium on Adaptive Dynamic Programming and Reinforcement Learning (**ADPRL**)*, Honolulu, USA, 2017.

[1] **George P. Kontoudis**, Minas Liarokapis, Agisilaos G. Zisimatos, Christoforos I. Mavrogiannis, Kostas J. Kyriakopoulos, "Open-Source, Anthropomorphic, Underactuated Robot Hands with a Selectively Lockable Differential Mechanism: Towards Affordable Prostheses," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Hamburg, Germany, 2015.

Referred Workshop Publications (W)

[2] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Path-Based Sensors: Will the Knowledge of Correlation in Random Variables Accelerate Information Gathering?" *IEEE International Conference on Robotics and Automation (ICRA), Workshop on Communication Challenges in Multi-Robot Systems: Perception, Coordination, and Learning*, London, UK, June 2023.

[1] **George P. Kontoudis**, "Scalable Multi-Robot Active Exploration," *Robotics: Science and Systems (RSS), Pioneers Workshop*, New York City, USA, June 2022.

Referred Abstract Publications (A)

[1] Mykola Mazur+, **George P. Kontoudis**, Anthony Petrella, "A Wireless Sensor Platform For Distal Extremity Movement Evaluation" *Orthopedic Research Society (ORS) Annual Meeting*, Phoenix, AZ, February 2025.

Theses (T)

[3] **George P. Kontoudis**, "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration," *Doctoral Dissertation, Virginia Tech*, USA, December 2021.

[2] **George P. Kontoudis**, "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation," *Master Thesis, Virginia Tech*, USA, December 2018.

[1] **George P. Kontoudis**, "Design and Development of an Underactuated, Anthropomorphic Robot Hand," *Diploma Thesis, National Technical University of Athens*, March 2016. *(in Greek)*

Technical Reports (TR)

[3] **George P. Kontoudis**, Daniel J. Stilwell, "Fully Decentralized, Scalable Gaussian Processes for Multi-Agent Federated Learning," *arXiv preprint, arXiv:2203.02865*, 2022.

[2] **George P. Kontoudis**, Minas Liarokapis, Agisilaos G. Zisimatos, Christoforos I. Mavrogiannis, Kostas J. Kyriakopoulos, "How to Create Affordable, Anthropomorphic, Light-Weight Prosthetic Hands," *Control Systems Lab, National Technical University of Athens,* Athens, Greece, October 2015.

[1] Agisilaos G. Zisimatos, Minas Liarokapis, Christoforos I. Mavrogiannis, **George P. Kontoudis**, Kostas J. Kyriakopoulos, "How to Create Affordable, Modular, Light-Weight, Underactuated, Compliant Robot Hand," *Control Systems Lab, National Technical University of Athens,* Athens, Greece, January 2015.

SPONSORED RESEARCH PROJECTS

External (Current)

- "Gaussian Processes that Embrace Heterogeneity for Multi-Sensor Sampling Mission," *National Science Foundation, Robust Intelligence*. PI: Michael Otte. Co-PIs: Pratap Tokekar, **George P. Kontoudis**. $1,000,000. Duration: 09/2026–08/2029.
- "Development of Next-Generation Dynamically Adaptive Orthoses for Distal Extremity Injury," *Henry M. Jackson Foundation for the Advancement of Military Medicine*. PI: Anthony Petrella. Co-PIs: Carl Frick, Andrew Petruska, **George P. Kontoudis**. $850,062. Duration: 07/2023–07/2027.

TALKS & PRESENTATIONS

- "Distributed Quantum Gaussian Processes for Multi-Agent Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026. **[Oral Presentation]**
- "Federated Gaussian Process Learning via Pseudo-Representations for Large-Scale Multi-Robot Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026. **[Oral Presentation]**
- "Learning to Act Together: Distributed Gaussian Processes for Scalable Multi-Robot Planning and Control," *Cyprus University of Technology*, Limassol, Cyprus, 2026. **[Oral Presentation]**
- "Learning Underwater Communication Maps for Large-Scale Multi-Robot Systems," *AOPE Seminar, Woods Hole Oceanographic Institution (WHOI)*, Woods Hole, USA, 2026. **[Oral Presentation]**
- "From Information to Dynamics: Optimal Kinodynamic and Informative Path Planning," *Robotics Seminar, CU Boulder*, Boulder, USA, 2025. **[Oral Presentation]**
- "Adaptive Informative Path Planning under Uncertainty via Gaussian Processes," *American Control Conference (**ACC**), Workshop: Data-Driven and Risk-Aware Control for Safety-Critical Autonomous Systems*, Denver, USA, 2025. **[Oral Presentation]**
- "Affordable Adaptive Robot Hands for Grasping and Manipulation," *NSF RET, Colorado School of Mines*, Golden, USA, 2025. **[Oral Presentation]**
- "Decentralized Informative Path Planning with Gaussian Processes," *NSF FRR-NRI PI Meeting*, Baltimore, USA, 2024. **[Poster Presentation]**
- "Optimal Kinodynamic Motion Planning and Informative Path Planning ," *Computer Science and Robotics Seminar, Colorado School of Mines*, Golden, USA, 2024. **[Oral Presentation]**
- "Decentralized Federated Learning using Gaussian Processes," *IEEE International Symposium on Multi-Robot and Multi-Agent Systems (**MRS**)*, Boston, USA, 2023. **[Oral Presentation]**
- "Adaptive Exploration-Exploitation Active Learning of Gaussian Processes," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Detroit, USA, 2023. **[Oral & Poster Presentation]**
- "Closed-Form Active Learning Using Expected Variance Reduction of Gaussian Process Surrogates for Adaptive Sampling" *American Control Conference (**ACC**)*, San Diego, USA, 2023. **[Oral Presentation]**
- "Decentralized Gaussian Process Learning for Exploration with Multi-Robot Systems" *Maryland Robotics Center Research Symposium*, College Park, USA, May 2023. **[Oral Presentation - Invited Talk]**
- "Scalable Exploration-Exploitation Active Learning of Gaussian Process Surrogates for Adaptive Sampling" *Maryland Robotics Center Research Symposium*, College Park, USA, May 2023. **[Poster Presentation]**
- "Multi-Robot Adaptive Sampling using Decentralized Gaussian Processes," *International Symposium on Distributed Autonomous Robotic Systems (**DARS**)* , Monbeliard, France, November 2022. **[Poster Presentation]**
- "Scalable Multi-Robot Active Exploration," *Robotics: Science and Systems (**RSS**), **Pioneers Workshop***, New York City, USA, June 2022. **[Spotlight Presentation]**
- "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration" *Bradley Department of Electrical and Computer Engineering, Virginia Tech*, Blacksburg, USA, December 2021. **[PhD Defense]**
- "Decentralized Nested Gaussian Processes for Multi-Robot Systems," *IEEE International Conference on Robotics and Automation (**ICRA**)*, Xi'an, China, 2021. **[Virtual Presentation]**
- "Online, Model-Free Motion Planning in Dynamic Environments: An Intermittent, Finite Horizon Approach with Continuous-Time Q-Learning," *American Control Conference (**ACC**)*, Denver, USA, 2020. **[Rapid-Interactive Presentation]**
- "A Comparison of Kriging and Cokriging for Estimation of Underwater Acoustic Communication Performance," *ACM International Conference on Underwater Networks and Systems (**WuWNet**)*, Atlanta, USA, 2019. **[Oral Presentation]**
- "An Adaptive, Humanlike Robot Hand with Selective Interdigitation: Towards Robust Grasping and Dexterous, In-Hand Manipulation," *IEEE-RAS Intern. Conference on Humanoid Robots (**Humanoids**)*, Toronto, Canada, 2019. **[Poster Presentation]**
- "Robust Kinodynamic Motion Planning using Model-Free Game-Theoretic Learning," *American Control Conference (**ACC**)*, Philadelphia, USA, 2019. **[Oral Presentation]**

- "A Compliant, Underactuated Finger for Anthropomorphic Hands," *IEEE/RAS-EMBS International Conference on Rehabilitation Robotics **(ICORR)***, Toronto, Canada, 2019. **[Poster Presentation]**
- "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation," *Department of Mechanical Engineering, Virginia Tech*, Blacksburg, USA, December 2018. **[Master's Defense]**
- "Evaluation Strategies of Adaptive, Anthropomorphic Robot Hands for Dexterous In-Hand Manipulation: Early Results," *National Institute of Standards and Technology **(NIST)***, USA, 2018. **[Invited Talk]**
- "Open-Source, Anthropomorphic, Underactuated Robot Hands with a Selectively Lockable Differential Mechanism: Towards Affordable Prostheses," *IEEE/RSJ International Conference on Intelligent Robots and Systems **(IROS)***, Hamburg, Germany, 2015. **[Oral Presentation]**

SERVICE ACTIVITIES

Editorial Service

- Associate Editor, IEEE Robotics and Automation Letters (RAL) *2025–present*
- Associate Editor, IEEE International Conference on Robotics and Automation (ICRA) *2025*
- Area Chair, Robotics: Science and Systems (RSS), Pioneers *2025, 2026*

Grant Proposal Review

- National Science Foundation (NSF) Reviewer *2024, 2026*

Conference Committees

- Program Committee Member, Australasian Conference on Robotics and Automation (ACRA) *2024*
- General Chair, Robotics: Science and Systems (RSS), Pioneers Workshop *2023*
- Program Committee Member, Conference on Artificial Intelligence (AAAI), Student Abstract and Poster *2023, 2024*
- Online Platform Chair, Conference on Robot Learning (CoRL) *2022*

Peer Review

- **Journals**: IEEE Transactions on Neural Networks and Learning Systems (TNNLS) · IEEE Transactions on Robotics (TRO) · IEEE Transactions on Automatic Control (TAC) · Automatica · IEEE Transactions on Automation Science and Engineering (TASE) · IEEE Robotics and Automation Letters (RAL) · IEEE Access (ACCESS) · IEEE Control Systems Letters (CSL) · IEEE Robotics & Automation Magazine (RAM) · Autonomous Robots (AURO) · IEEE Journal of Oceanic Engineering (JOE) · IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMCA) · IEEE Transactions on Cybernetics · IEEE Computational Intelligence Magazine (CIM) · Frontiers in Robotics and AI (FROBAI) · Frontiers in Artificial Intelligence (FAI) · Journal of Optimization Theory and Applications (JOTA) · International Journal of Advanced Robotic Systems (IJARS)
- **Conferences**: IEEE International Conference on Robotics and Automation (ICRA) · IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) · Robotics: Science and Systems (RSS) · American Control Conference (ACC) · IEEE Conference on Decision and Control (CDC) · International Symposium on Distributed Autonomous Robotic Systems (DARS) · IEEE Conference on Control Technology and Applications (CCTA) · International Conference on Ubiquitous Robots (UR) · IEEE International Conference on Automation Science and Engineering (CASE) · IEEE-RAS International Conference on Humanoid Robots (Humanoids) · European Control Conference (ECC) · IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob) · Mediterranean Conference on Control and Automation (MED)

University Service

- Graduate Admission Committee Member *2024–present*
  Mechanical Engineering Department, *Colorado School of Mines*
- Graduate Admissions Committee Member *2026*
  Robotics Program, *Colorado School of Mines*
- **Judge**: Mines Graduate Research and Discovery Symposium (GRADS) - S '24, S '25 · Mines Capstone Design Senior Showcase - S '24, S '25 · Mines Undergraduate Research Symposium - F '24, S '25, S '26

RESEARCH ADVISING

PhD Students

- Aaron Fanganello, Robotics, Colorado School of Mines *2025–present*
  Awards: 2026 SMART Scholar
- Dave Fobar, Robotics, Colorado School of Mines *2025–present*
- Meet Gandhi, Robotics, Colorado School of Mines *2025–present*
  Publications: C-[20]
  Awards: 2025 Mechanical Engineering Fellowship, 2026 AAMAS Doctoral Consortium, 2026 AAMAS Student Scholarship
- Mykola Mazur (co-advised with Anthony Petrella), Mechanical Engineering, Colorado School of Mines *2024–present*
  Publications: A-[1], C-[18], J-UR[1]
- Sanket Salunkhe, Robotics, Colorado School of Mines *2024–present*
  Publications: C-[19]
  Awards: 2026 AAMAS Student Scholarship
- Joshua Netter (*mentored with Kyriakos G. Vamvoudakis*), Georgia Institute of Technology *2020–2023*
  Publications: C-[12], J-[8]

Master's Students

- Leo Paparoni (co-advised with Anthony Petrella), Colorado School of Mines *2026–present*
- Olivia Franklin, Colorado School of Mines *2026–present*
- Remi Jones, Colorado School of Mines *2026–present*
- Fahad Asat, Colorado School of Mines *2026–present*
- Elijah Paiz, Colorado School of Mines *2025–2026*
  Next Position: Strength Analyst, United Launch Alliance (ULA)
- Jude Ortiz, Colorado School of Mines *2025–2026*
- Alkesh K. Srivastava (*mentored with Michael Otte*), University of Maryland *2022–2023*
  Publications: C-[14], W-[2], C-[21]
  Next Position: PhD student, Temple University
- Zirui Xu (*mentored with Kyriakos G. Vamvoudakis*), Georgia Institute of Technology *2018–2020*
  Publications: C-[7], BC-[1], J-[5]
  Next Position: PhD student, University of Michigan

Undergraduate Students

- Graden Gentry, Colorado School of Mines *2026–present*
- Analissia Wise, Colorado School of Mines *2026–present*
- Audrey Smith, Colorado School of Mines *2025–present*
  Awards: 2026 FIRST and SURF Scholar, 2027 MURF Scholar
- Matthew Smith, Colorado School of Mines *2025–present*
- Arael Anaya, Colorado School of Mines *2025–present*
  Awards: 2025 IEEE CASE Travel Grant, 2025 SURF and MURF Scholar
- Evan Pacic, Colorado School of Mines *2024–present*
  Awards: 2025 FIRST and MURF Scholar, 2026 SURF Scholar
- Olivia Franklin, Colorado School of Mines *2025–2026*
  Next Position: MS student, Colorado School of Mines
- Remi Jones, Colorado School of Mines *2024–2026*
  Awards: 2nd poster place 2024 Fall Undergraduate Research Symposium, 2026 Undergraduate Research Scholar Distinction
  Next Position: MS student, Colorado School of Mines
- Jude Ortiz, Colorado School of Mines *2025*
  Next Position: MS student, Colorado School of Mines
- Elijah Paiz, Harvey Scholar, Colorado School of Mines *2024–2025*
  Awards: 2nd poster place 2024 Fall Undergraduate Research Symposium
  Next Position: MS student, Colorado School of Mines
- Ross Hodson, SURF and MURF Scholar '24, Colorado School of Mines *2024–2025*
  Next Position: MS student, Oregon State University

Visiting Students

- Ethan Brown, FIRE Scholar '25, Brigham Young University — *2025*
  Next Position: MS student, Brigham Young University

PhD Students - Dissertation Committee Member

- Thong Quoc Huynh, Robotics, Colorado School of Mines — *2026–present*
- Bailey Hopkins, Robotics, Colorado School of Mines — *2026–present*
- Eugene Hamzezadeh, Robotics, Colorado School of Mines — *2025–present*
- Matthew Stanley, Mechanical Engineering, Colorado School of Mines — *2024–present*
- Seth Stewart, Robotics, Colorado School of Mines — *2025*

MS Students - Thesis Committee Member

- Yee Shen Teoh, Computer Science, Colorado School of Mines — *2024–2025*

PhD Students - Qualifying Exam Committee Member

- Linda Knudsen, Mechanical Engineering, Colorado School of Mines — *Spring 2026*
- Jiuyi Xu, Robotics, Colorado School of Mines — *Spring 2026*
- Haozhixiang Li, Mechanical Engineering, Colorado School of Mines — *Fall 2025*
- Kazi Rifat Bin Rafiq, Mechanical Engineering, Colorado School of Mines — *Fall 2024, Spring 2025*
- James Verheyden, Mechanical Engineering, Colorado School of Mines — *Fall 2024*
- Eugene Hamzezadeh, Robotics, Colorado School of Mines — *Spring 2024*
- Seth Stewart, Robotics, Colorado School of Mines — *Spring 2024*